PRINCE AGYEI TUFFOUR

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EDUCATION

Oregon State University Corvallis, OR
M.S. in Mathematics — Computational Mathematics, Numerical Linear Algebra, Statistical Modeling

EXPERIENCE

dynaConnections Corporation
Software QA Engineer *Present*

- Designed **automated test suites** across API and data pipeline layers, reducing production defects by 40% and accelerating release cycles for an enterprise analytics platform.
- Built **monitoring dashboards** to track data quality, system performance, and model drift across staging and production environments, cutting incident response time by 30%.
- Validated **ML model outputs** in partnership with cross-functional data science and engineering teams, ensuring integration reliability in user-facing applications.
- Authored **technical documentation** of testing assumptions, data lineage, and result interpretation for stakeholders across engineering and product teams.

Akomapa Health Foundation
Chief Technology Officer (Non-Profit) *Present*

- Led development of a **patient management platform** for non-communicable disease tracking across multiple countries, translating complex healthcare business requirements into analytical solutions.
- Architected data infrastructure with **Python, SQL, and Azure** to support patient record analytics, EDA workflows, and clinical outcome dashboards serving cross-functional clinical teams.
- Managed a distributed engineering team, coordinating **sprint planning, code reviews, and deployment pipelines** to deliver features on schedule across time zones.

Cita Marketplace
Machine Learning Engineer Intern

- Built and deployed **recommendation models** (collaborative filtering, content-based) in Python (scikit-learn, pandas, NumPy), improving product discovery metrics for a 500K+ user dataset.
- Conducted **exploratory data analysis** on user behavior data, extracting actionable patterns that informed A/B test design and feature prioritization.
- Developed **data preprocessing pipelines** to clean, transform, and feature-engineer raw transactional data, reducing model training time and improving prediction accuracy.

PROJECTS

Stormlog — GPU Memory Profiling Toolkit Open Source · PyPI

- Engineered a **production-grade profiling library** for PyTorch and TensorFlow that monitors GPU/CPU memory allocation and growth to **optimize inference latency and cost** during training and deployment. Published on PyPI (8 releases, 357+ commits, CI/CD, unit tests, CLI, TUI).
- Diagnosed a **PyTorch memory leak on Apple MPS** backend, publishing a technical article on framework-level debugging across CUDA, ROCm, and MPS accelerator stacks.

Veridex — RAG Evaluation Framework GenAI

- Built an end-to-end **RAG pipeline** with document chunking, embedding generation (OpenAI/Sentence-Transformers), and retrieval via **FAISS and ChromaDB vector stores**, enabling context-grounded Q&A over domain-specific corpora.
- Implemented an **LLM evaluation suite** measuring faithfulness, answer relevance, and context precision to detect **hallucination**, achieving 92%+ grounding accuracy on benchmark datasets using automated scoring and human-in-the-loop validation.

Randomized SVD for Scalable PCA Research

- Implemented **randomized matrix factorization** (NumPy/SciPy) to compute rank-k approximations of large matrices, achieving near-identical variance explained to full SVD at **10x+ speedup** on high-dimensional image and feature extraction tasks.

CodeExplain — AI Documentation Generator GenAI

- Built an **LLM-powered tool** that auto-generates code documentation via structured prompt engineering and a **GenAI pipeline**, with output quality evaluation to minimize hallucination.

ML from Scratch

- Implemented core **ML algorithms from first principles** (regression, SVMs, decision trees, neural networks, ensembles) in Python to deepen model **explainability and bias auditing** capabilities.

TECHNICAL SKILLS

ML & Data Science: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers, pandas, NumPy, SciPy, Matplotlib
GenAI & LLMs: RAG pipelines, vector stores (FAISS, ChromaDB), prompt engineering, LLM evaluation, parameter-efficient tuning, hallucination detection
Engineering & Cloud: Python, SQL, Spark/PySpark, Docker, Kubernetes, AWS, Azure, GCP, Git, CI/CD, REST APIs
Methods: A/B testing, EDA, model evaluation, statistical modeling, numerical linear algebra, optimization, monitoring & drift detection